

Universal Card Shuffler

Final Report

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Executive Summary

Trading Card Games and Board games do not have an automatic way to shuffle their cards in a safe and quick manner. Traditional hand shuffling techniques are time-consuming and often not effective. The objective of this project is to design an automatic card shuffler capable of shuffling any type of card in any card game or board game.

Technical problems of the project include designing a mechanism capable of adjusting to any card dimensions, creating a system that is safe for the cards so that they will not be damaged, developing a method of shuffling which ensures high levels of randomness, and ensuring the final price of the machine falls within the budget of the target demographic.

After reviewing responses from our early ideation end user survey, the key objectives and performance specifications of this project were developed. Firstly, and most importantly, the solution must be safe for the cards. In 100 shuffle cycles, the target is for no cards to be damaged. Secondly, the speed should aim to be under 30 seconds to beat out most people in hand shuffling. Thirdly, the randomness of the shuffle should end up being better than regular hand shuffling. Our method for determining randomness will be with variation distance. Please refer to the Customer Requirements and Engineering Design Specifications section of the report for more information. The other 3 targets are price, to be under \$75 so the general public can afford it, size, as preferred by many people in the target audience survey to be within 6 inches cubed, and noise, to be under 50 decibels so it doesn't disturb the game.

The tools that will be used in this project include Solidworks for CAD design, Bambu Labs for 3d printing and prototyping, stm32 microcontrollers for controlling sensors and motors, and iterative testing with user feedback to perfect the final product.

The final design of choice is the "Two Tower System" (Appendix A). This system uses 2 flywheels positioned opposite each other to spin out cards from both sides into the middle. In the middle is a catcher that catches the cards into a neat pile, and a splitter which divides the cards back into 2 piles. After all the cards are shuffled, a lift raises the centerpiece up and dispenses the 2 piles back into the 2 towers to allow for more shuffle cycles.

An early prototype was assembled to test the feasibility of the system in shuffling cards to great success. A link to the video can be found in Appendix B. The prototype is a simple barebones

version of the shuffler that was modelled off a shuffler on Amazon. The prototype was made to assess the reliability of the proposed method of shuffling, to test microcontroller programming, and to gain an estimate of the approximate size of the final product. Compared to the cheap amazon shuffler, our prototype was a lot larger, designed to accommodate cards of various sizes, and featured an automated method of stopping the system with an ultrasonic sensor to detect the presence of cards.

After the early prototype, new electrical components were chosen to downsize the product and increase precision and accuracy. Additionally, CAD for the final product was produced (Appendix A) to introduce the recirculation system. These CADs were then 3d printed and iterated through to adjust for 3d printing tolerances.

The next steps for this project are to integrate the electrical components with the mechanical components to produce a fully working card shuffling system. After the full assembly, the plans are to patent the product and send it to local game stores to gain valuable end user feedback on the system and continue to improve it.

Nomenclature and Glossary

LGS – Local Game Store

MTG – Magic: The Gathering

TCG – Trading Card Game

Riffle – A Riffle Shuffle, a type of shuffle commonly used in casinos

1. Introduction and Background

The Universal Card Shuffler project aims to design and develop an innovative, adjustable card shuffling device capable of handling cards of various dimensions, including sleeved cards commonly used in trading card games (TCGs). Card games, ranging from traditional playing cards to popular TCGs like Magic: The Gathering, Pokémon, and Yu-Gi-Oh! are enjoyed by millions worldwide. A critical aspect of these games is ensuring a fair and randomized shuffle to maintain game integrity. However, manual shuffling can be time-consuming, physically taxing, and prone to errors, particularly for players with large decks or limited dexterity. Additionally, many TCG players use protective card sleeves to preserve the condition of their cards, which increases the card dimensions and complicates shuffling with standard devices.

Existing automatic card shufflers are typically designed for standard poker or bridge-sized cards and lack the flexibility to accommodate the diverse sizes and thicknesses of sleeved TCG cards. This limitation creates a gap in the market for a versatile shuffling solution that can adapt to various card dimensions while ensuring a thorough and unbiased shuffle. Our capstone design team seeks to address this challenge by developing a universal card shuffler that offers adjustable mechanisms to handle cards of different sizes, with a particular focus on supporting sleeved cards used in TCGs.

The objective of this project is to create a user-friendly, efficient, and reliable device that enhances the gaming experience for TCG enthusiasts and casual card players alike. By incorporating adjustable components and robust shuffling mechanisms, the shuffler will cater to a wide range of card games, ensuring compatibility with both sleeved and un-sleeved cards. This compatibility will be tested through repeated shuffle cycles on the machine. In 100 cycles, no cards should be damaged, and the randomness should be at an acceptable level with a variation distance of 0.334.

In addition to the above objectives, the system should be safe for the cards, shuffle faster than by hand, be a portable size, be quiet, and also be an affordable price.

This progress report outlines the design process, including problem definition, research, and preliminary design concepts, as well as our progress towards the final finished product.

2. Existing Products in the Card Shuffler Market

The market for card shufflers includes a range of products, from inexpensive, consumer-grade devices to high-end, casino-quality machines. These shufflers vary significantly in price, functionality, and compatibility with different card types, particularly sleeved cards used in trading card games.

Low-cost card shufflers, typically priced between \$10 and \$30, are widely available through retailers like Amazon, Walmart, and Target. An example would be the Feierya Card Shuffler (Appendix C), which is priced at \$15. These relatively cheaper devices are primarily designed for standard poker or bridge-sized playing cards and are often battery-operated or manually cranked. Most are not designed to handle sleeved cards, which are thicker and wider due to protective sleeves commonly used in TCGs. User reviews and community feedback, such as posts on Reddit's r/boardgames and r/magicTCG, frequently report issues with jamming when sleeved cards are used, as the mechanisms are too narrow or lack the precision to accommodate the added thickness. Additionally, low-cost shufflers can damage cards, especially delicate TCG cards, due to aggressive rollers or poor alignment, leading to bent edges or torn sleeves. Even for un-sleeved cards, these shufflers often provide suboptimal randomization, with some models performing a basic weave shuffle that fails to meet the randomization standards required for competitive TCG play.

Higher end shuffler models such as those used in casino shufflers tend to be much more expensive with models such as the One2six-OTS Single deck and Multideck Shuffler (Appendix D) being priced at \$7000. These higher end machines allow for greater randomization while being able to protect cards from damage as well. However, even these shuffles are not able to appeal to the TCG market as they are not able to adjust for various card dimensions. The high cost and limited flexibility of these devices create a gap in the market for our product to enter.

3. Codes and Standards

The card shuffling device will include various powered components to be sold as a consumer product, thus requiring adherence to a few standards. For electronic safety standards, the product will adhere to the requirements set forth by FCC Part 15 Subpart B to regulate

electromagnetic interference from products if a device operates at a frequency of 9 kHz or higher [1]. Similarly for the power sources, any chosen battery will adhere to the lithium ion testing regulations per UN DOT 38.3 regulations [2][3]. As is industry standard for most consumer products, the device would also undergo and receive UL certification prior to any mass production [4].

Additionally, since the shuffling device is intended to be used for various card games in a competition and recreational applications, there are some fairness and shuffling rules which are considered during the design of the device. For traditional card games, Gaming Labs International has an outlined set of rules and standards for casino-style card games in terms of how cards must be shuffled and dispensed [5]. Most importantly, for Magic: The Gathering, there is an outlined documentation of tournament and comprehensive rules that mention constraints on how shuffling must be done. As the players of these two games comprise most of our target audience, the rules set forth in their respective documentation will be used as a benchmark standard for the shuffling device.

4. Stakeholder Analysis

There are four main stakeholder groups identified for such a device as shown in Table 1 consisting of Magic: The Gathering (MTG) players, general trading card game competitive players, the average card game players, and lastly unknown intention uses. The stakeholders are organized and evaluated based on two factors: influence and importance. As seen in Figure 2, the stakeholder with the most importance is MTG players as a product like this is a desire for all players and would be the device of its kind for the popular game. The unknown player was included to acknowledge the use of this card shuffling device for a completely different, yet similar application. This stakeholder obviously would hold the least influence out of the rest, however it was still included to be conscious of all design features so that if the opportunity or use of such a device in an unexpected market exists it could be used. c

Table 1: Stakeholder

Stakeholder	Interests	Impact/Effect	Importance	Influence
magic card game players	Play Magic with decks consisting of sleeved cards	insight into the needs of target market	High influence, High Importance	Main users, largest consumer base
TCG card game competition players	play in competitions where sleeved cards are used, and shuffling is important	provide feedback for specific things to be mindful for competition style use	High influence, High Importance	official channels for promotion of such a device to the target audience
average card game player	people who play a variety of different card games (classic, pokemon, magic, etc.)	provides wants and uses of an average person	High influence, Low Importance	provides feedback for making the device "universal" for all uses
Unknown Game Players	Users who may need this device for a different kind of use or unplanned card game	provide insight into other applications of the product	Low influence, High Importance	Feedback on tailoring device to unintended applications

Analysis

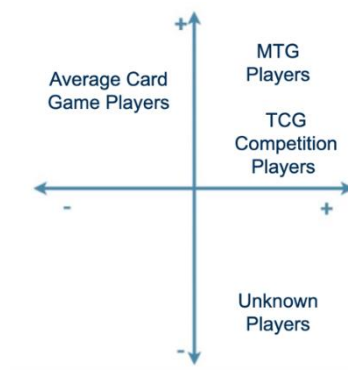


Figure 1: 2x2 Stakeholder Matrix

5. Market Research

With the primary stakeholders identified, the next step was to perform initial market research on the end users. A targeted survey was distributed via Google Forms (Appendix E) to the Georgia Tech Magic: the Gathering Club. This group represents the target clientele for this product due to their high use of card shufflers with varied size decks and non-standard playing cards. The survey collected 39 responses about gameplay frequency, shuffle time, satisfaction with randomness, and spending on card games as well as preferred device size, desired features, and market potential. Secondary research was also conducted on online sources to estimate the size of the gaming market and the competitive landscape of other potential products that are in this niche.

The global gaming accessories market is valued at between \$10-\$14 billion, with card shufflers forming a small segment of between \$100 million and \$200 million [6]. The board game market, including card games like Magic the Gathering, was valued at \$14.37 billion in 2024 and is

expected to reach over \$16 billion in 2025 [7]. The card shuffler niche can be valued at anywhere between 5 million and \$20 million with a current high demand for shufflers that can handle sleeved cards and large decks.

Those that responded to the survey represent the core of this market as frequent card game players, with 47% of respondents playing card games weekly and 13% playing daily. 74% of respondents spend between \$50 and \$200 on card games annually. This money can be used for deck boxes, playmats, dices and other equipment used during gameplay. Most people were willing to pay between \$20 and \$50.00 for an automatic card shuffler, with 68% preferring the price tag to be under \$20. 24% of people were willing to pay up to \$30 and only 8% were willing to pay up to \$50. It is possible that since only students were surveyed that this low price tag range reflects the budget of typical college students And could suggest that the general market would be willing to pay slightly more. A price tag over \$50 is still significantly less than professional card shufflers like the One2six OTS Shuffler that retails for around \$7000. However, it is still more than the FEIERYA Card shuffler that is only \$15. While the cheap shuffler is extremely cost efficient, its shuffling style is very primitive, and it is also known to damage cards. A price tag of around \$50 could be justified to ensure card protection as well as satisfactory randomness of shuffles.

The main go-to market strategy for this product would be to sell through e-commerce platforms to reach student gamers as well as gamers all around the world. Marketing could be performed at major tournaments as well as on any streams online of gameplay. Sales in gaming stores would also be very beneficial as it would allow potential customers to interact with samples before making a purchase. However, this is a secondary sales method due to the reach of e-commerce.

With an estimated 200,000 Magic: the Gathering players globally and 47% of these people playing weekly, a very conservative 5% market penetration would produce around 470,000 shuffles annually. Since this card shuffler will be universal and can be used for other types of games as well As for standard card games, It is possible that this product could achieve 1,000,000 shuffles worldwide per year. This goal was absolutely achievable because the product that is being designed fills a gap in the market. There is currently no other card shuffler on the market that both aims to protect cards as well as perform multiple shuffling cycles. There's also a large disparity between retail everyday card shufflers in the range of \$20 to \$60

and the very high end card shufflers that range from high \$500 to up to \$10,000. A product that can bridge the gap between these two segments of the market and provide the best aspects of both could perform incredibly well.

6. Customer Requirements and Engineering Design Specifications

The results of the market research and especially the Potential User Survey aided in the creation of a House of Quality (Appendix F), to help analyze and determine the customer requirements and engineering design specifications (Appendix G).

In the survey, 84% of those who responded prioritized no damage to sleeved cards whatsoever. 55% of those who responded when a device that is compact in size and 13% mentioned wanting a device that is quiet in operation. This means that the device must have slots that can accommodate thicker than normal cards, as well as have a delivery mechanism that does not damage the cards in any way. The demand for multiple shuffling cycles also requires the need for a microprocessor for control over shuffles in order to achieve full satisfaction with randomness every time and address the 55% of people who responded to the survey that were dissatisfied with their current shuffling method. The desire for this device to be small and portable means that it must be battery powered, and it must be durable to withstand many changes in location. Affordability is also an important requirement for the device with the target audience being an average consumer, and as such must have a shelf cost of around \$75 or less. The device must also be of a portable size so it can be easily transported around. In the survey, a target size was identified to be within 6 inches cubed.

Additionally, the product and operating life were also considered requirements, despite not being correlated with survey results. Requirements such as these are non-negotiable for machines being sold to a consumer. The target of this project was a product life of ~10+ years (1000+ uses) which is industry standard for most commercial products, and an operating life of at least 6 hours between each charge, which is the maximum length an average tcg tournament can go for.

Several constraints were also identified for the product, mainly within the manufacturing and electronic components of the device. Given the importance of the device's ability to shuffle

cards of all geometries, the machining tolerances of parts used specifically to adjust the input thickness of cards is important. Also, the capability of whatever microcontroller module used can be limited if many motors/sensors are powered as a module may fail or glitch if it is heavily used.

With all of this information put into the House of Quality (Appendix F), the most important engineering requirement was identified to be choice of shuffling mechanism. This is because it would have a very large impact on several high weight customer requirements such as reliability, card safety, and ability to adjust for card dimensions. The next most important requirements were manufacturability and product lifespan.

A set of engineering specifications (Appendix G) were then compiled to encompass all the aforementioned information and organized based on category of general, physical, mechanical, electrical, or performance. One of the most notable specifications is the randomness of the shuffle, which will be measured by performing tests with cards in a predetermined order. The result of each test shuffle will be repeated and compared against the expected order to ensure consistent performance. The tests will also record the variation distance to quantify the measured distance of the shuffled deck from its original order before shuffling. Together these two methods can statistically verify the randomness of its operation.

Given the robustness of such a device for validation of all the individual specifications, the primary method of validating these are met will be through repeated use testing of the device.

7. Design Concept Ideation

Card Shuffler Functions & Subfunctions (See Appendix H: Function Tree and Appendix I: Morphological Chart)

The Universal Card Shuffler must fulfill the main functions of shuffling cards, ejecting shuffled cards, and being adjustable for universal compatibility. The shuffling function can be broken down into five subfunctions: shuffling method, counting cards, splitting the deck, flipping misoriented cards, and recirculating cards. Adjustability can be separated into two subfunctions: adjust for card length/width and adjust for thickness. For the most part, ejecting cards is its own subfunction, so there is no need to elaborate further. Some subfunctions, such as flipping misoriented cards and recirculating shuffled cards, are less important than others

and may be worth ignoring for simplification and future prototyping. To better understand the functions discussed below, check the morph chart.

The most important subfunction to design for the Universal Card Shuffler is how it will shuffle cards, as this will determine what other subfunctions are necessary; other subfunctions also have this effect. The primary three shuffling methods considered as a solution were firstly, a wheel that holds all the cards and shuffles using an algorithm, secondly, a dispenser design that alternates two sides of the split deck, and thirdly, a design that uses chambers to separate the cards into two or more groups and stack them together.

If the wheel is selected, deck splitting and recirculation subfunctions can be eliminated entirely since the wheel has a place for each card, and there is no need to shuffle more than once to achieve a desired level of randomness.

As mentioned, both the alternating stack method and the chambers method of shuffling require the deck to be split. One solution to this is using forks like a forklift to bisect the cards and move the top layer to the side; a similar solution is using a wedge in the same fashion, but the wedge would allow the top layer to slide down to the side instead of requiring a lift to move the cards. Another solution is using powered rollers that spin in one direction for the first half of the deck and the other for the second half, splitting the cards into two piles. One more uses a gate that opens and closes off one of two directions, alternating between the two for each card.

Again, for some of the aforementioned split mechanisms, the shuffler needs to be able to detect the cards. One possible solution is an IR or light sensor similar to those used in money counters; the mechanism counts cards by counting up every time a card passes over the sensor. Another solution is a rotating wheel with slots that uses an encoder to count cards one at a time; this mechanism is derived from the end roller on some money counters that stacks bills individually, but the system is more passive as it does not do the actual counting. One more uses a scale to estimate the number of cards by dividing the average weight of the cards by the measured weight. A potential workaround is to simply make an estimate on where to split the deck and adjust the positioning manually.

Most shuffling methods here require the deck to be recirculated for a better shuffle than competitors. The four main solutions to card recirculation are a conveyor belt, an elevator, or a rack and pinion. Each of these mechanisms serves the same purpose: moving the shuffled cards from the bottom of the system back to the top. Another potential solution is having the

user manually move the cards back to the top of the shuffler, but this is no different from how most other shufflers solve this.

An important part of performing a sufficient shuffle is making sure the cards are all oriented correctly. There are two methods for flipping misoriented cards: an extending and retracting overhang that would allow any backward cards to flip while falling to a platform underneath and a flipper arm that picks up the card and rolls it over. This subfunction would only be possible if a sensor system is used to count the cards, and that system would need to be able to analyze each card and flag any backward cards quickly. The shuffler's ability to flip misoriented cards would be a function that further sets the Universal Shuffler apart, but it would further complicate the design. Flipping misoriented cards overall is a much less important feature than the rest, as it should not be difficult for the user to insert their cards correctly.

The shuffler's ability to accommodate different playing cards is the core of what makes it "universal," so it is crucial for its dimensions to be adjustable. For the length/width of the card, the card tray could be manually adjustable, or ultrasonic sensors could be used to adjust the system automatically. For the thickness or to make sure only one card is moved at a time for certain subfunctions, one side of the card tray could be equipped with a manually adjustable sliding gap, a spring-loaded roller that pinches the card, ensuring only one gets through at a time, or another ultrasonic sensor attached to the gap that automatically adjusts gap depending on the measured thickness of the card. It is almost possible to design this system with a "one size fits all" approach, simplifying the design.

The final function the shuffler needs to perform is ejecting the shuffled cards. Two methods to accomplish this are ejecting a card tray with a user-controlled number of cards and outputting the cards one at a time.

With all functions and solutions determined, 3 potential final designs were developed.

Potential Integrated Concepts

The first design is the one with 2 towers which dispenses cards into the middle alternating between the halves(Appendix J). This design is similar to a more common and inexpensive card shuffling product. However, those existing products have a few problems worth addressing, such as their inability to accommodate cards outside standard playing cards, flip misoriented cards, or recirculate them for a better shuffle. The concept uses a distance measuring sensor to

detect the presence of cards, rollers in alternating directions to split the deck, an elevator to recirculate the cards, a manually adjustable slider for dimensional adjustment, and a tray that can shift its position to split the shuffled deck into 2 for recirculation. This tray will be adjusted manually to split the deck evenly.

The major advantage of this design is that shuffling should be relatively quick, and the user would be able to control how many times the deck recirculates. A major disadvantage is that the formula for shuffling cards is not as complex and for an optimum shuffle, multiple recirculations would be required, increasing the time of the shuffle. Manufacturing-wise the system should also be relatively cheap as although it has a few parts, they are all very simple and will not require any complex processes. The system uses a few moving parts but not many of them interact with the cards themselves, making the system safe for the user.

Another integrated concept uses a wheel to shuffle the cards (Appendix K). Instead of shuffling by physically changing the order of the cards, this design instead uses an algorithm to randomize the cards and output cards in that randomized order. It also needs to keep track of what slots in the wheel are filled for if the user needs more cards or adds ejected cards back into the system. This concept is similar to an existing casino-based system, which only works with a single deck of cards and spits out one card at a time.

As stated, the wheel also solves more problems than just shuffling cards, so fewer parts are required for this concept, simplifying its overall design. Although this is a more streamlined design, it is more complicated and requires more electronics and moving parts than the previous concept. These moving parts can potentially cause jams in the machine, which can be hazardous for the cards. Additionally, the wheel must have an exceptionally large diameter in order to store a large amount of cards, making it the largest design out of our potential concepts. This complex wheel would also increase costs in production as manufacturing is not simple.

A more impractical concept incorporates a scale to count the cards (Appendix L), a wedge to separate the cards, chambers to shuffle, an overhang to flip misoriented cards, an elevator for recirculation, a manually adjustable tray, and outputs cards one at a time. The chambers system pushes the cards into two or more chambers (potentially up to 10) with trap doors underneath each. Once all the cards are sorted, the doors open in a random order, and the cards are dropped into a shuffled pile underneath.

Similar to the wheel design, this design would also require an algorithm to randomize the order of the cards, but the mechanism for shuffling is simplified, so there is no need to keep track of individual cards and when they are dispensed. This design is unlike anything else on the market, which could help it stand out, but its complex configuration may need to be reconsidered in future designs. This design uses the least amount of moving parts, making it very quiet but due to its many chambers, it increases the points at which cards could get jammed. This idea of shuffling is also very simple and would produce the worst randomness out of the 3 designs.

8. Concept Selection and Justification

Below is table showing how each of our 3 potential final system performs against the engineering requirements taken from the House of Quality. Each specification is rated on a scale of 1 to 3 with 3 being the best design for the requirement and 1 being the worst. Additionally, each specification has an attached weight, taken from the house of quality. Requirements that would fundamentally be the same for all 3 designs were removed from the table below.

Table 2: Evaluation of designs

Specification	Weighting	Design 1	Design 2	Design 3
		Two Tower	The Wheel	Conveyer and Chamber
Speed	0.08	2	1	3
Randomness (Quality)	0.06	3	3	1
Price	0.1	2	1	3
Size	0.07	2	1	2
Card Safety	0.22	3	2	1
Product Lifespan	0.12	2	1	1
Manufacturability	0.18	3	2	1
Sum:		2.12	1.35	1.26

Therefore, our final design of choice is the Two Tower design.

9. Industrial Design

The universal card shuffler prioritizes the industrial design considerations of manufacturability, functionality, and physical ergonomics. From early market research it was clear that protecting cards, the size of the device, and simplicity of operation would be key considerations. The design placed emphasis on implementing solutions based on customer needs. The tower design accommodates sleeved cards and allows users to adjust the thickness of their cards. This addresses the user request of minimal damage to cards. The tower design highlights physical ergonomics as cards are easily placed and run through a cycle of seven shuffles with minimal user input. The design was influenced by the principal of matching form to function and keeping manufacturing and assembly simple.

The visual language was chosen to reduce the cognitive load of use. The two-tower design makes it clear where cards are supposed to go. Thus, operation of this product is as complex as existing card shuffling devices. Our device makes it clear where cards should be placed and a simple button input is required for starting the sequence. It was necessary to compromise between the aesthetic appeal and manufacturability for the prototype. Materials selected for production will be ABS plastic as it will be durable, but light enough for transportation. This also allows the product to be designed for manufacturability as pieces can be snap fit. A final production version would likely include a outer casing as well as matte finish on the ABS components.

Manufacturing prototypes did not include branding, and the focus was not on aesthetics but function. However, in future work simple branding would be incorporated into the injection molded parts as well as clearly marked labels to indicate where cards will go and how to perform simple operation. The color of the device would likely remain black or another neutral color as to not be obtrusive in a card game tournament setup.

10.Detailed Technical Analyses, Experimentation, and Design Performance Prediction

In order to validate the design multiple tests were performed using the functional prototype in order to simulate the performance of the finished product. The results of these tests were then

compared with the desired technical design specifications and performance targets of the team.

The first test centered around the shuffling mechanism of the product as it is the central function of the Universal Card Shuffler. The goals of the test were to produce a 0% damaged card rate, a 0% jam rate, and a single cycle shuffle time of less than four seconds. The four second target was the only test that was finite in length as the two 0% targets were consistently being evaluated during all usages of the device.

The product proved to be able to shuffle an entire deck of Magic the Gathering card, which is more cards than a standard playing card deck in 3.5 seconds. This metric when extrapolated out the full intended functionality of the device would produce 21 seconds of total shuffling over the course of one usage of the device leaving another 39 seconds for all other functions to achieve the target of one usage taking less than 60 seconds. See Appendix B for link to a video.

In addition to the success of this test, during all tests of the device during the semester not a single card was damaged, and the product did not jam once achieving both the 0% damage rate target as well as the 0% jam rate target.

The other performance test that was done on the prototype was the randomness test. This test was rather difficult to construct as desired randomness is a subjective metric. However, it was necessary to test this in some way due to the requests of product's future customers that responded to the end user survey. Therefore, it was decided to use the numeric system of a playing card deck in order to verify the relative movement of cards after one full usage cycle.

A standard playing card deck was rearranged so that the cards were arranged to be grouped by value in ascending order. The functional prototype was then run 6 times to simulate the full usage of the final product. After each shuffle cycle, the deck was cut in half and placed back in the towers just as the product would do. After the 6th shuffle the deck was then checked for its order as shown in Figure 2.



Figure 2: Deck Order Examination After Full Usage

After 10 tests, it was found that 9 of them produced a final deck that had no instances of cards of adjacent value being directly next to each other. This confirms satisfactory randomness of the shuffling mechanism and algorithm as at minimum 90% of the time the shuffle spreads the cards out adequately from their original order.

11.Final Design and Mockup

The CAD of the final design is shown in full below:

Figure 3

The final design functions in 5 separate steps. Firstly, the user splits the deck into 2 even piles to put on either side of the machine. A distance sensor under both piles detects the presence of the cards which causes the machine to start operation. Next, a fly wheel located at the edge of either tower spins cards into the middle one at a time. The wheels operate on a delay of 30ms so one wheel starts spinning before the other to ensure cards do not collide with each other. As

they are shuffled into the middle, an initial catcher catches the cards and naturally keeps them all together in a nice pile. After all the cards have been distributed to the center, a tray positioned on a rack and pinion below the catcher initially starts at an offset to one side where half the cards will fall into. This tray will then move to the other side and cause the rest of the cards to fall into the other side. In the next step, the tray centers itself and the lift elevates the centerpiece until it is above the two towers. In the final step, 2 small pushers on the tray spread apart and push the piles back into the tower.

The associated bill of materials for the design is below:

Table 3: Bill of Materials

Function	Part name	Pros	Cons	Price
Dispensing Cards	Mini Micro N20 Gear Motor DC 3V-6V x6	Smallest motor available on the market at the cheapest price	Requires an awkward input voltage, may need additional components to regulate voltage and current	\$1.5 x 6
Sensing remaining cards	VL6180X	Very precise, can start measuring from 0cm compared to other sensors which require a gap of at least 2cm	Programming a time of flight sensor is more difficult than simple infrared or ultrasonic sensors	\$1.6
Microcontroller Unit	STM32F103C8T6	Very small, very cheap	Much more difficult to program compared to an Arduino with lots of libraries	\$1.5
Battery	Anker PowerCore+ mini	Power bank makes everything simple, usb connects directly to the MCU	Not customizable	\$8.25
Gear Rack	20 Degree Pressure Angle Gear Rack 48 Pitch	Precise tolerance and quality, flexible material	Material is strength is not great	\$3.89 x 3
Rubber Wheel	Rubber Wheel, Black, Soft 1-5/8" Diameter x 3/4" Wide	Exact dimensions and high quality	Could still damage cards	\$2.32 x 2
Plastic Miter Gear	Plastic Miter Gear, 48 Pitch, 18 Teeth	Precise tolerance and quality	Price	\$12.98 x 4

Plastic Gear	20 Degree Pressure Angle Plastic Gear Round Bore, 48 Pitch, 18 Teeth	Precise tolerance and quality	Price	\$5.00 x 4
			Total Cost	\$102.01

All Mechanical components were purchased from McMaster-Carr and all electrical components were purchased off Amazon.

Although this final price is above the target manufacturing costs of \$50, a large portion of this price is from the expensive mechanical components purchased from McMaster-Carr. If the machine were to be mass-produced, this price could see a significant decrease as we move to buying in bulk.

Current progress on the final product includes finished 3d prints of all the parts, as well as basic circuitry being completed. Circuitry currently includes soldering all the connections together and wiring all the motors to their respective motor controllers and onto the main microcontroller. Using the Stm32IDE, a simple full cycle of the shuffle was also programmed as a starting point. With the next steps being to incorporate the distance sensor to properly control the starting and stopping of the code.

12. Manufacturing

Manufacturing Process

Throughout every stage of development, the Universal Card Shuffler was engineered explicitly with 3D printing as a core consideration. Most parts were designed with simple shapes and minimal overhang, to ensure they could be quickly and easily printed with nominal material used for support. This approach significantly streamlined the prototyping and reiteration process as parts could be freely edited or reconfigured to meet specific goals and reprinted with minimal effort. It also enabled rapid mechanical functionality and fit testing, allowing for continuous improvement without the delays and costs associated with other manufacturing methods.

The first official prototype was almost entirely 3D printed and lacked many of the mechanisms included in the final design. It was a minimal, bare-bones iteration focused solely on the main

card shuffling mechanism, serving primarily as a proof of concept. The remainder of the design was composed of and assembled using miscellaneous components, including salvaged parts and spares from previous projects. While far from polished, this initial version of the shuffler played a crucial role in validating the shuffling mechanism and informing the direction of future iterations. Going forward, the shuffler would be designed to incorporate off-the-shelf components to promote standardization, improve reliability, and simplify assembly.

The core design established in the first prototype became the foundation for all subsequent iterations of the shuffler, remaining relatively unchanged, with many of the remaining components and features built around it. This consistent foundation allowed for modular development, enabling new mechanisms to be added or adjusted without altering the core functionality, further simplifying the reiteration process.

As the project moves toward mass production, 3D printing will be phased out in favor of injection molding, as the shuffler's simple design, particularly with components such as the card holders and lift tower, lends itself well to the injection molding process. This transition will enable increased production efficiency, reduced manufacturing costs, and enhanced final product durability. To further reduce manufacturing costs, minimize the number of individual components, and lessen dependency on external suppliers, non-moving parts, such as the racks, could be integrated directly into the components they would be attached to in the current iteration. The gears could also be manufactured in-house, and given their standardized design and low strength requirements, they should be inexpensive and straightforward to produce.

Impact on Design and Production Methods

Since the current iteration of the shuffler was designed to be mostly 3D printed, design considerations extended beyond just its shape. It needed to be compatible with a wide range of filaments (similar to PLA), printable within typical tolerance ranges of consumer-grade 3D printers, and composed of parts that could fit within the build constraints of smaller printers. These considerations guaranteed that the parts could be printed on any available printer without issue.

The shuffler's parts are all rigid, avoiding mechanisms that would require the use of flexible or high-strength materials. By designing with exclusively standard thermoplastics like PLA in mind, the design was kept intentionally simple while ensuring broad compatibility with virtually all

consumer-grade 3D printers. Additionally, no individual 3D-printed component exceeds 9 inches in any dimension, contributing further to the print's broad compatibility.

Many of these 3D printed parts are interlocking and were therefore designed with carefully considered tolerances to ensure a consistent fit and alignment regardless of minor variations in printer calibration. As previously noted, all of the mechanical systems within the shuffler use standard off-the-shelf parts, many of which were sourced from McMaster-Carr (gears, racks, etc.), so the tolerances and dimensions of these standardized components were also taken into account during the CAD modeling process to ensure proper integration and reduce the need for post-processing. A key aspect of the design was finding a balance between creating a tight fit that would allow sourced parts to work properly and adding clearances that would allow for slight dimensional discrepancies. Fit tolerances range from 0.05 mm to 0.3 mm.

As stated, the Universal Card Shuffler was developed with 3D printing as a central consideration throughout its early design and prototyping phases. However, as the project transitions to mass production, fabrication will shift to injection molding. Factors such as packaging and storage requirements must also be addressed.

In addition to altering production methods, a casing will be designed around the shuffler to protect its internal mechanisms without compromising its functionality; this change significantly simplifies and improves the efficiency of both packaging and storage. For packaging, the change eliminates the need for additional protective materials and allows for packaging to be much more compact than it would be if the components were packaged unassembled.

A couple more things to consider with regard to storage are that the shuffler is mainly made of plastic and has a rechargeable lithium battery inside. The shuffler would need to be stored in a cool, dry environment to reduce the chance of the plastics warping and degrading and prevent the corrosion of electronics. It would need to be stored at 40-60% charge to minimize long-term capacity loss, and the packaging could potentially incorporate ESD protections. The packaging must also be designed to withstand stacking without significant deformation, ensuring the shuffler remains protected from stress during storage or transit.

13. Societal, environmental and sustainability considerations

The “Social Impact Assessment” methodology for this card shuffler follows the stages below where the team looked at the goal and scope, inventory analysis, and then generated an interpretation of these results.

I. Goal and Scope

Objective of Assessment	Design Function	Functional Unit	Lifecycle Stages Considered	Associated Activities
Assess the social impacts of the Universal Card Shuffler	Offers a randomized shuffle for cards of varying dimensions.	1 Universal Card Shuffler	Manufacturing	Injection Molding
				Product Assembly
			Product Use	Customer Use

II. Inventory Analysis

Product Lifecycle Stage	Stakeholder Group	Social Impact Category	Impact Indicators
Manufacturing	Workers	Health and Safety	# of injuries per year
Product Use	Consumers	Feedback Mechanisms	Rating for Product % Competition Acceptance

III. Interpretation of Results

The universal card shuffler overall does not have much of a positive or negative impact to human-well being in general. It is able to offer convenient shuffling for a niche consumer base of trading card game players that seek this product. A positive impact of this product would be that due to its relatively inexpensive production cost, it would have the ability to be made in the United States, and therefore offer the creation of employment. A possible negative social impact of this product could be the workplace hazards due to industrial machinery such as injection molding being used in large volume production. To mitigate negative impacts certain

steps would need to take place such as adherence to factory safety regulations. Furthermore to address the possible environmental impact due to excess scrap plastic through the production line, there will need to be a recycling method integrated that minimizes scrap from injection molding.

The chosen relevant product life cycles stages for this analysis are the manufacturing and product use stages. These are of the greatest importance for this product as we are extremely reliant on consumer feedback as well as manufacturing as this is a product that would have a production process. Based on these two important stages the targeted stakeholders became the customers and the worker at the production facility. With regards to consumers the best measure of this social impact includes ratings the product receives as well as its acceptance into the trading card game competition realm. As for impact on workers, the major focus relies in a safe workspace as this is extremely vital especially in an industrial settings such as the one the universal card shuffler would be manufactured in.

14. Risk Assessment, Safety, and Liability

A failure mode and effects analysis (FMEA) was completed to identify potential risks with the card shuffling machine as shown in Appendix M. The analysis assigned a risk priority number for each identified potential risk or function, based on three factors: severity, occurrence, and detection. The analysis indicated that the biggest failure modes lie in power consumption, machining tolerances, functioning kinematics, and control algorithms. Given that the product is a relatively small, enclosed machine, there weren't many specific safety issues taken into consideration outside the ones caused by the failure risks.

To mitigate the main risks, the shuffler was designed to have a low power consumption and a more simplified controls layout. The number of motors used was reduced by using accompanying mechanisms such as gears and levers to minimize any strenuous load that may cause electrical components to fail. A simple 18650 battery is used to power the system as well, reducing any risk of fire or hazards from occurring. By keeping everything in an enclosed system, there lie almost no safety concerns for the user, even if mechanisms were to fail.

15. Conclusion

To summarize, for the universal card shuffler project, we were able to identify a gap in the Trading Card Game market for a device capable of shuffling cards of any dimensions. A problem statement was created to address this need as well as detail our intentions as a team on how to solve this problem. End users were then surveyed to help develop the customer needs and engineering requirements for the project. A Function Tree and Morphological Chart were then made to determine the key functions required as well as how to achieve these functions. Out of these charts, 3 Potential designs were identified. These designs were all evaluated against the engineering requirements and 1 final design which best fit these requirements was selected.

Initial prototyping and early testing on the feasibility of performing a single cycle was evaluated and iterated until perfected. The designed machine can perform a shuffle equivalent to a riffle shuffle able in 4 seconds. The lift system was also prototyped and iterated with the other mechanisms until a working design was reached. The electrical wiring diagram of the mechanism was also designed and tested, ensuring proper control of all motors and sensors in communication with the microcontroller. The finalized prototype was then introduced and tested by potential customers for feedback. The team is currently worked very closely with the Magic the Gathering club at Georgia Tech, keeping them updated with progress and organizing potential times when the club may be able to test the prototype. The final shelf design and mass manufacturing of the card shuffling device was planned, as outlined in the above report.

16. Team Member Contributions

Joshua Wu—Gantt Chart, Engineering Requirements, Function Tree, Design 1 CAD, Report, Circuit building, programming, Final Design CAD, Prototyping

Ali Komilian—Engineering Specifications, Stakeholder Matrix, Customer Requirements, Morphological Chart, Design 2 CAD, Report, Prototyping, Final Design CAD

Sarfaraz Syed—Survey Creation, House of Quality, Specification Sheet, Design 3 CAD, Report, Prototyping, Final Design CAD

Alexander Lutz—Market Research, Presentations, Report, Editing, Prototyping, Final Design CAD, Expo Video and Poster

Devinn Ingram—User Needs and Design, Morphological Chart, Design 2 CAD, Design 3 CAD, Report, Prototyping, Final Design CAD, Fabrication Package

Angad Singh—Market Research, Design 3 CAD, Report, Social Impact Analysis (SIA), Prototyping, Final Design CAD, Expo Video and Poster

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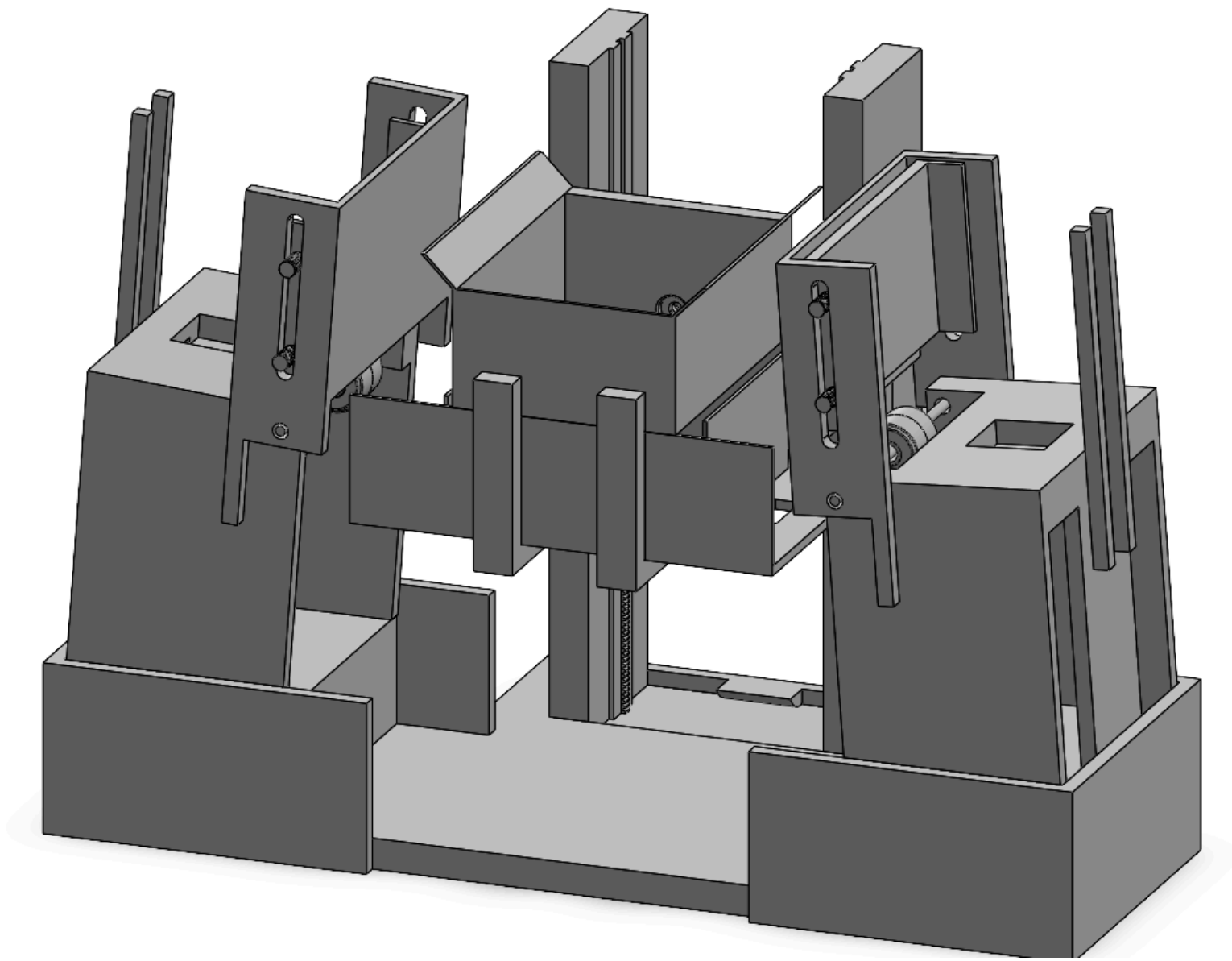
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Appendix

Appendix A: Final design



Appendix B: Video of prototype:

https://youtu.be/cs_fdS0fooY

Appendix C: Feierya Card Shuffler



Appendix D: One2six-OTS Single deck and Multideck Shuffler



Appendix E: End-User Survey

Timestamp	How often do you play a game requiring cards to be shuffled?	How long does it take you to shuffle a deck of cards you're playing with?	How often are you satisfied with the randomness of your shuffle?	Do you own any devices to help shuffle cards?	How much money do you spend a year on card games /board games	When you go to play in a local competition, what stuff do you bring?	If you were to have an automatic way to shuffle your cards on the market, how much would you pay for it?	If a machine were to exist, what would be the maximum size of the machine for you to be comfortable with bringing it around with you?	What are some key aspects you would want to see in the card shuffler? e.g. safe for your cards	Do you see a future of a machine like that becoming popular and becoming a staple in the table-top gaming community?
5/26/2025 17:53:12	Weekly	30-40 seconds	Always	No	\$150-\$200	play mat, decks, dice, life pad	Under \$20	maybe 2 deck boxes worth of size	safe for cards is a requirement I would also say decently light weight and not to loud	Yes
5/26/2025 18:04:17	Weekly	20-30 seconds	Sometimes	No	Under \$50		\$20-\$30	6" cube	Safe, can use sleeved and unsleeved cards	Yes
5/26/2025 19:14:47	Daily	Takes upwards of 40 seconds but I'm shuffling a 100 card deck	Always	No	\$200+	My decks, a playmat and dice	Under \$20	Has to be able to fit into a backpack with all of my other stuff	No change in the quality of my cards	No
5/26/2025 19:59:48	Weekly	30-40 seconds	Sometimes	No	\$200+	Decks boxes dice etc	\$30-\$40	Probably about the size of a deck box	Probably just a bare bones shuffler that maintains orientation	Yes
5/27/2025 7:28:38	Weekly	10-20 seconds	Always	No	\$200+	Playmat, dice, tokens, decks, sleeves	Under \$20	About the size of a deckbox	Safe for cards, can shuffle double-sleeved cards	No
5/27/2025 13:08:42	Weekly	40 seconds +	Always	No	\$50-\$100	Decks of cards, associated supplies for contract bridge	\$20-\$30	easily fits into most bags and can withstand minor jostleing	consistently works, battery powered, useful for different card deck sizes	No
5/27/2025 13:20:57	Monthly	30-40 seconds	Always	No	\$200+	Playmat, Deckbox, Dice, Tokens.	Under \$20	No bigger than an Ultimate Guard Boulder deck box	Does not damage/crease corners on card sleeves, quiet.	No
5/27/2025 13:26:55	Rarely	30-40 seconds	Never	No	Under \$50		Under \$20			Yes
5/27/2025 13:29:54	Weekly	10-20 seconds	Sometimes	No	\$100-\$150		\$20-\$30		Speed and randomness. Existing card shufflers are slower than manual and do not randomize properly. Inherently slow as you need to put the cards in, which can be weird. Would need to do it in <5s (52-card deck) for me to use it	Yes
5/27/2025 13:41:54	Monthly	Under 10 seconds	Sometimes	No	\$100-\$150		\$20-\$30		Useful for sleeved cards	Yes

5/27/2025 13:58:57	Daily	10-20 seconds	Sometimes	Yes	\$50-\$100	Nothing	Under \$20	Pocket size	Don't bend the cards	Yes
5/27/2025 14:33:41	Rarely	40 seconds +	Sometimes	No	Under \$50		Under \$20	Pocket		Yes
5/27/2025 14:37:22	Monthly	10-20 seconds	Sometimes	No	Under \$50	nothing	Under \$20	handheld	won't fold the cards	Yes
5/27/2025 15:21:57	Rarely	20-30 seconds	Sometimes	No	Under \$50	I don't participate	Under \$20	Slightly bigger than a wallet	Buttons to select options	No
5/27/2025 16:52:44	Rarely	30-40 seconds	Sometimes	No	\$100-\$150	I don't partake in these events	Under \$20	size of the deck	actually makes shuffling easier	No
5/27/2025 16:53:36	Rarely	10-20 seconds	Sometimes	No	Under \$50	n/a i don't play in local competitions	\$20-\$30	no larger than 3"x6"x6" or so.	replacable batteries	No
5/27/2025 17:03:15	Monthly	10-20 seconds	Sometimes	Yes	Under \$50		\$20-\$30			
5/27/2025 18:09:56	Weekly	20-30 seconds	Sometimes	No	Under \$50		Under \$20	The size of the deck of cards probably	Doesn't result in my cards being super bent or folded, Doesn't take longer than I do to shuffle	No
5/27/2025 19:56:25										
5/27/2025 22:04:00	Weekly	20-30 seconds	Sometimes	No	Under \$50		\$40-\$50			
5/27/2025 22:06:56	Monthly	20-30 seconds	Sometimes	No	Under \$50		Under \$20	copy machine sized		No
5/27/2025 22:28:29	Monthly	30-40 seconds	Always	No	Under \$50	Cards	Under \$20	Size of iphone	Lightweight	Yes
5/27/2025 23:01:16	Rarely	Under 10 seconds	Sometimes	No	\$50-\$100	Cards	Under \$20	Deck of cards	Not hurt the cards	No
5/27/2025 23:07:19	Monthly	Under 10 seconds	Sometimes	No	\$50-\$100	Just poker chips	\$20-\$30	I'd want it small	It work and not mess up my cards	Yes
5/28/2025 0:24:59	Monthly	10-20 seconds	Always	No	Under \$50		Under \$20			No
5/28/2025 1:15:27	Rarely	30-40 seconds	Sometimes	No	Under \$50		\$30-\$40			
5/28/2025 2:52:44	Monthly	10-20 seconds	Sometimes	No	\$50-\$100	Nothing	Under \$20	The size of a tissue box	Won't damage my cards, good shuffling, and fast	Yes
5/28/2025 3:22:24	Monthly	40 seconds +	Sometimes	No	\$50-\$100		\$30-\$40	Pocket sized		Yes

5/28/2025 10:37:30	Monthly	20-30 seconds	Always	No	Under \$50	poker chips	\$20-\$30	maybe like a little desktop sized thing	safe for cards safe for kids	Yes
6/1/2025 15:08:24	Weekly	40 seconds +	Sometimes	No	\$150-\$200	Card box, playmat, dice	\$30-\$40	6x6x6”	Can accommodate double sleeved cards, safe for cards, not super loud	Yes
6/3/2025 15:51:30	Weekly	20-30 seconds	Sometimes	No	\$200+	my mind	\$20-\$30	small	consistently works, doesn't break, and is actually truly shuffling them	Yes
6/3/2025 16:16:16	Rarely	20-30 seconds	Always	No	\$50-\$100	I don't go to local competitions	\$20-\$30	A little bit bigger than a deck of cards	It spins by itself and you can put in how many people are playing in the game	Yes
6/6/2025 12:19:09	Weekly	Im adhd so sometimes i just get distracted and br shuffling	Sometimes	No	\$200+	Deck box and mat	\$20-\$30	12" x 6"	Doesnt damage sleeves, needs to be quiet (playing card shufflers have a loud whiny motor sound)	Yes
6/6/2025 12:19:10	Weekly	10-20 seconds	Sometimes	No	\$200+	Mat, backpack deckbox, trade binder	Under \$20	No bigger than a sidewinder deckbox	Gentle on cards, very smooth and safeguard to not shuffle if card is sideways etc	No
6/6/2025 12:20:33	Weekly	30-40 seconds	Sometimes	No	\$50-\$100	My Stuff	\$20-\$30	About the size of a basic ultra pro deck box, maybe a little bigger	DOES NOT BEND/WARP, preferably not loud	Yes
6/6/2025 13:05:20	Daily	10-20 seconds	Sometimes	No	\$200+	Backpack, deckbox, deck, mat, dice, pen, notepad, extra sleeves. Those are the bare minimum in my bag usually	\$30-\$40	Hard to describe the answer but no bigger than a few inches bigger than 2 halves of a deck placed side by side	Safety is the most important consideration. Ease of use(loading/unloading cards). The sounds it makes(not too loud)	
6/6/2025 13:42:20	Weekly	10-20 seconds	Sometimes	No	\$200+	Mat, deck boxes with cards inside, binder, bookbag to carry it all	Under \$20	Slightly larger, and a 2-3 inches thicker, than a standard iphone	Safety for cards would be #1. They're all in sleeves so the shuffler couldn't damage or snag sleeves at all or else it might make accidentally “marked” cards	No
6/6/2025 15:55:07	Daily	10-20 seconds	Always	No	\$200+	Cards, playmat, deckboxes, sleeves, dice, counters, token cards	\$20-\$30	A double-sided deckbox (Twin Flip-&-Tray for the larger end if it also had storage for accessories)	Built in deck box, compartments for various accessories (dice, counters, coins)	Yes
6/11/2025 17:50:22	Weekly	20-30 seconds	Always	No	\$200+	/	\$50+	30cm/9 Inch	Safe for cards, big decks and dealing to player positions	Yes

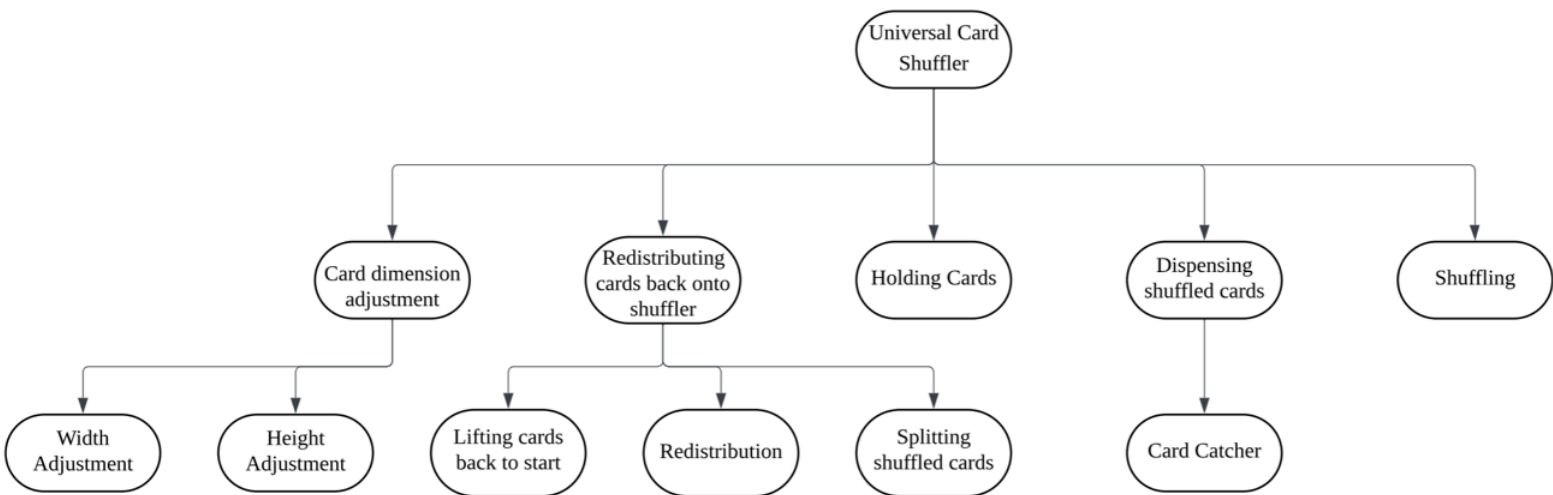
Appendix F: House of Quality

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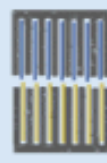
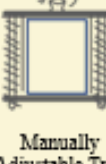
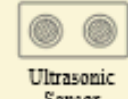
Appendix G: Specification Sheet

			For: Universal Card Shuffler	Issued:	5/20/2025	
			Specification			
No.	Date	D/W	Requirements	Responsible	Source	How Validated
General						
1	5/20/2025	D	no damage to cards	design team	online sources for material in moving system that reduces damage to components	repetitive testing with various types of cards and sleeves to analyse if damage is sustained
8	5/20/2025	W	<\$75 price range	design team	expected shelf price	get manufacturing estimates for bulk production
10	5/20/2025	D	extended product life	design team	expected number of uses during lifetime	cyclic fatigue testing until failure of the machine to determine approximate use life
Physical Characteristics						
2	5/20/2025	D	Adequate base dimensions	market research team	average size of devices on the market	user feedback and conformity to the industry norm
9	5/20/2025	W	strong, durable material, resistant to common wear	market research team	material properties of machines in the market	chosen design material is compared to those used in similar products
Electrical						
3	5/20/2025	D	Controlled card output velocity	design team	testing for adequate speed range	motor speeds will be varied until an usable output speed is found to launch the cards out
4	5/20/2025	D	Count Cards	design team	identify way to read count of loaded cards	testing to validate the output count based on the input number of loaded cards
5	5/20/2025	W	Central control system	design team	advanced user input functions similar to higher end machines in the market	testing of input system and associated mechanisms to support smooth function and user interface
Mechanical						
5	5/20/2025	D	Adequate card tolerance system	design team	design of a mechanism that can be varied	tested with various card thicknesses and geometries to validate "universal" function
6	5/20/2025	D	Split deck	design team	identify way to split the deck at the halfway point	repetitive testing to show proper splitting of the deck
7	5/20/2025	D	Elevator System	design team	identify mechanisms which can be used to efficiently circulate the cards	repetitive testing to show proper function of the mechanism
Performance						
8	5/20/2025	D	Ability to complete a full shuffle cycle	design team	card shuffling standards	test with a pre determined order of cards to see if the final output order is as expected
11	5/20/2025	W	Flip misoriented cards	design team	identify and flip a card in the wrong orientation	testing with a rigged deck to see if the device can recognize and fix the error

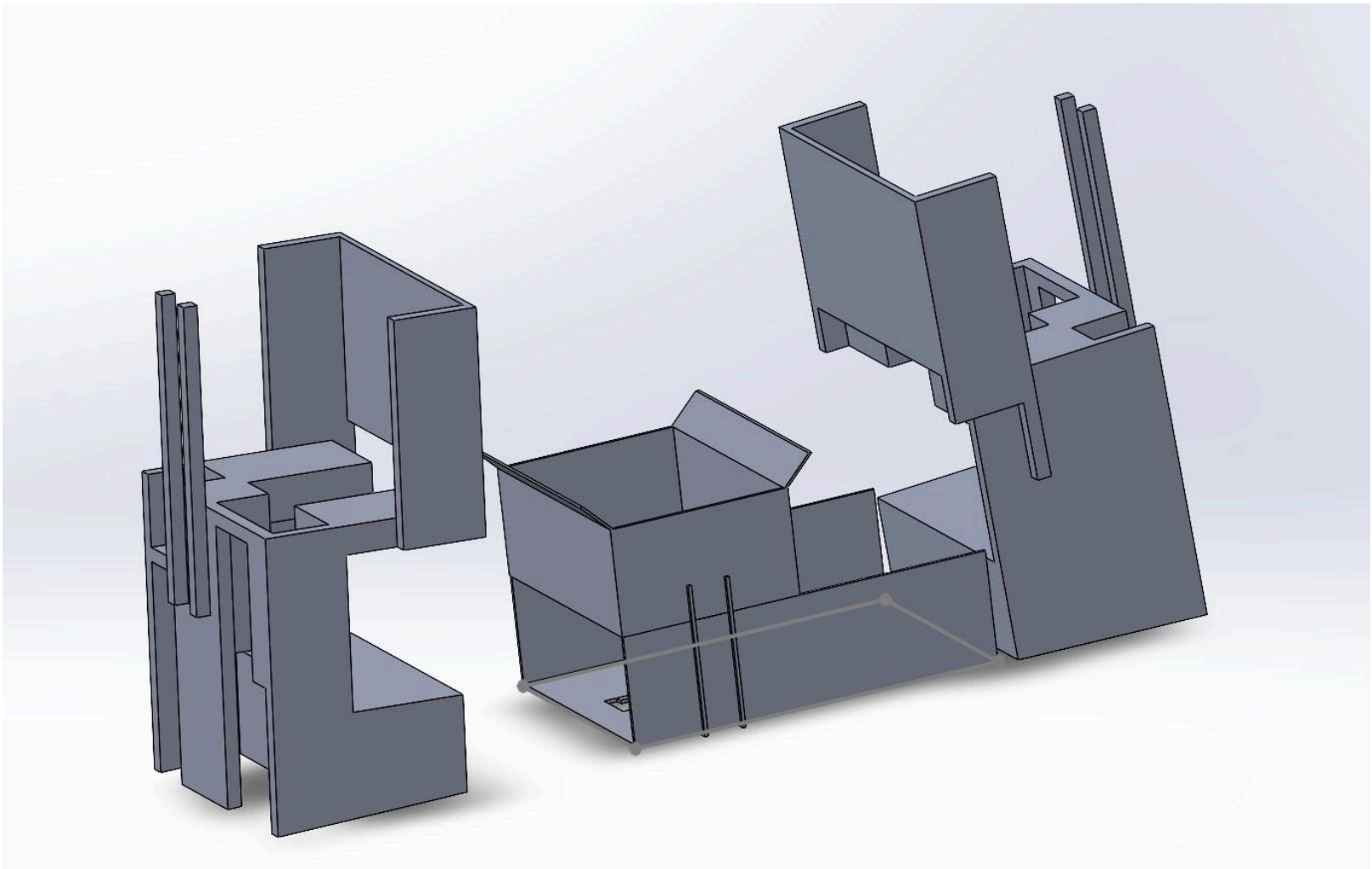
Appendix H: Function Tree



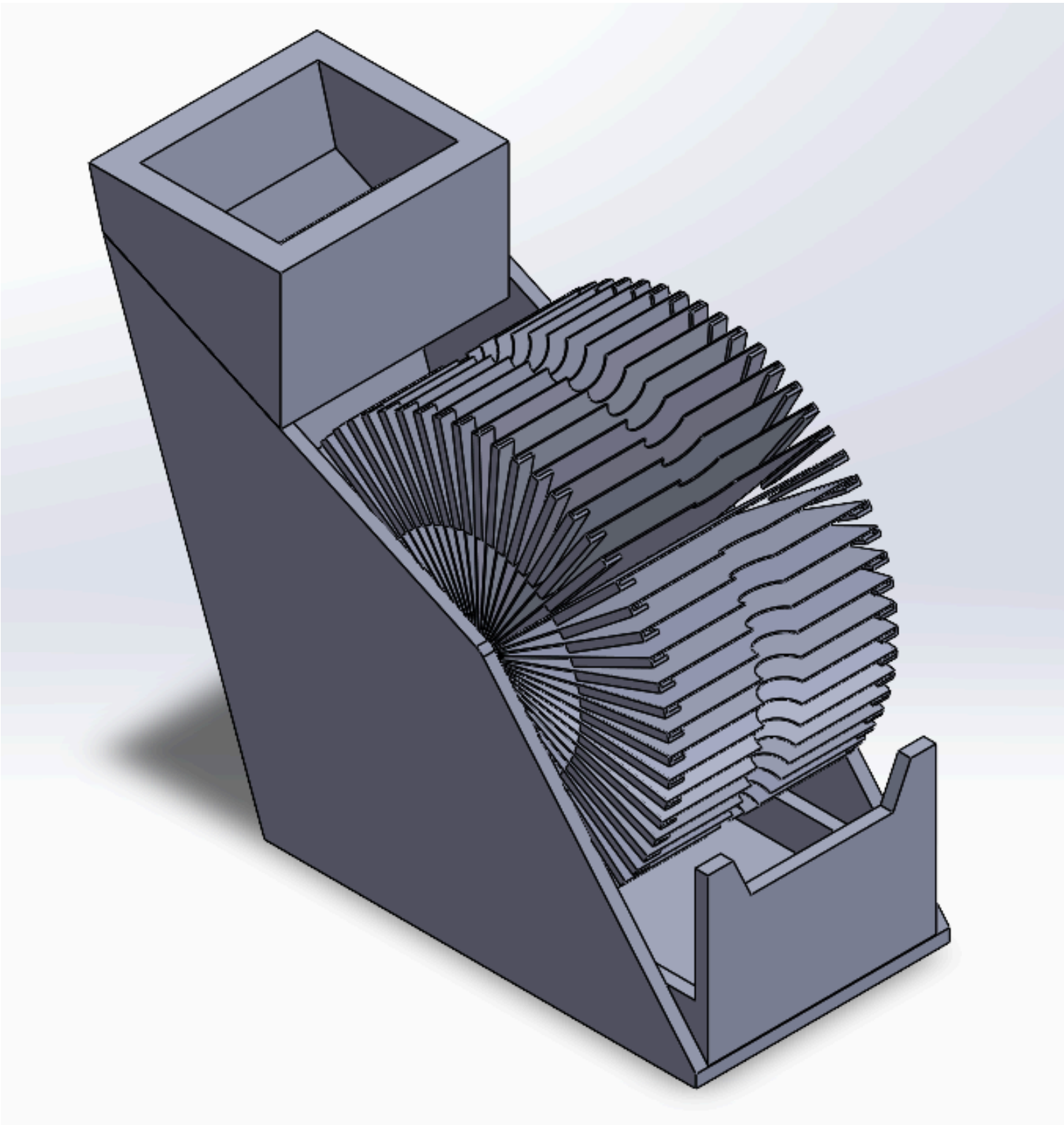
Appendix I: Morphological Chart

Sub-Functions	Solutions					
Shuffle Cards	 Wheel	 Alternating Stacks	 Chambers			
Count Cards	 Counting Roller	 IR Sensor	 Scale			
Split Deck	 Forklift	 Wedge	 Rollers	 Gates		
<u>Recirculate</u> Shuffled Cards	 Conveyor Belt	 Elevator	 Rack & Pinion	 User Moves Cards Back to Top	 Vacuum Grippers	
Flip Misoriented Cards	 Overhang	 Flipping Arm				
Adjust for Length/Width	 Manually Adjustable Tray	 Ultrasonic Sensor	ONE SIZE FITS ALL			
Adjust for Thickness	 Sliding Gap	 Spring-loaded Roller	Ultrasonic Sensor	ONE SIZE FITS ALL		
Eject Shuffled Cards	 Card Tray	 One at a Time				

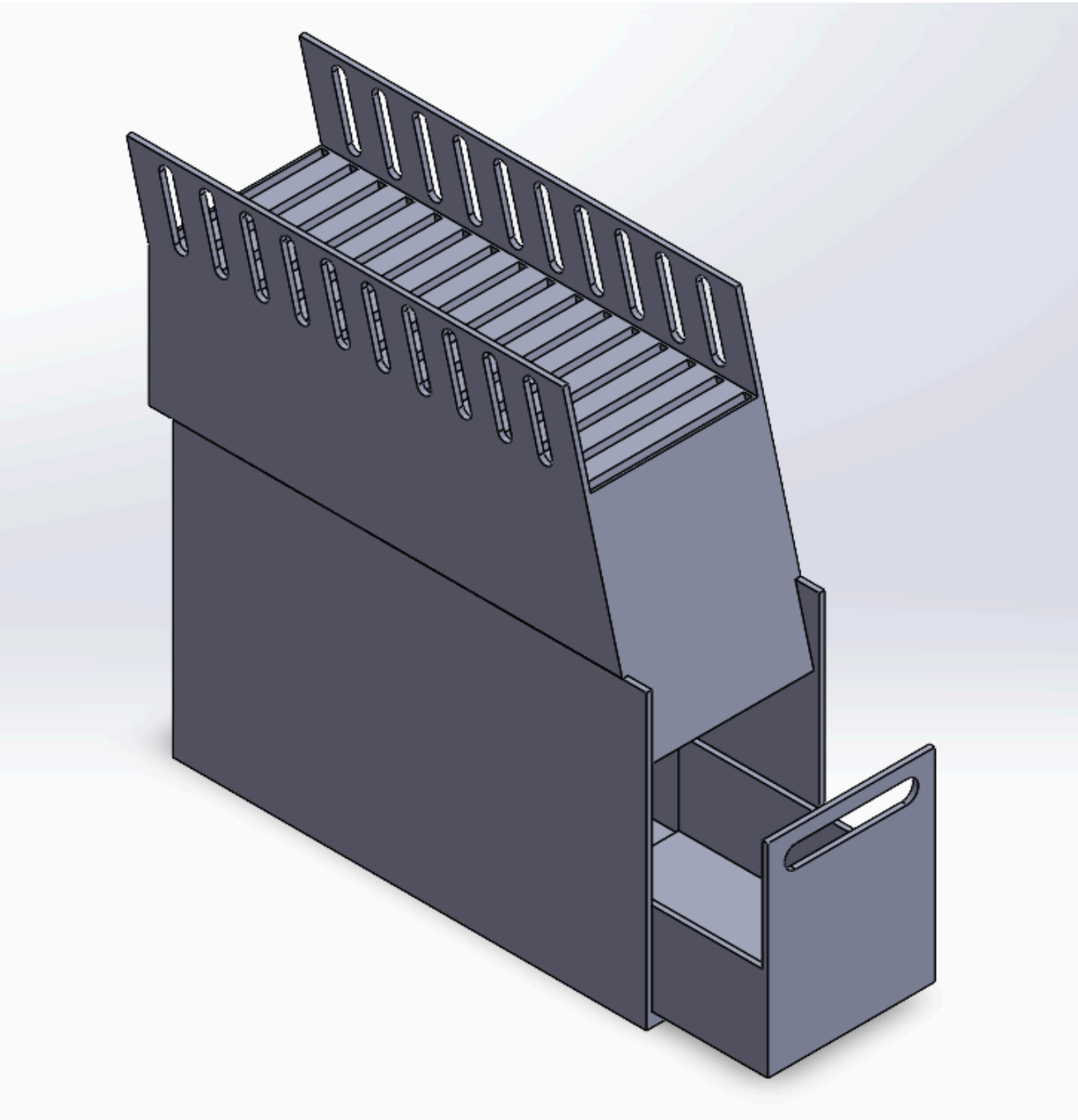
Appendix J: Two Tower Design



Appendix K: Wheel Design



Appendix L: Conveyer and Chambers



Appendix M: FMEA Analysis

Item/Function	Potential Failure Mode	Potential Effects of Failure	Potential Causes	Current Controls	Severity (S)	Occurrence (O)	Detection (D)	RPN
Power consumption	Excessive power draw from micro controller	Overheating, component damage	Poor circuit & wiring, motor oversizing & overworking	Power budget analysis, load testing	7	4	4	112
Fasteners & load-bearing materials	Bolt/shear failure	Structural collapse, jamming	Undersized bolts, poor material choice and tolerancing	Material specs, material certification	6	2	3	36
Fatigue of load-bearing parts	Crack growth, fracture, shearing	Machine failure, minor injury risk	Cyclic stress, poor fastening, intense loads	Fatigue calculations, material choice	6	2	3	36
Stability during operation	Tipping, vibrations	In accurate shuffle, machine damage	Poor base design, imbalance, high vibration	Center of gravity, vibration testing	7	3	4	84
Kinematics of card transport	Misalignment, jams	Shuffle errors, downtime	Design tolerances too tight/loose	Kinematic modeling, prototyping	7	4	4	112
Dynamic load analysis	Excessive forces during operation	Bearing wear, motor overload	High acceleration profiles	Dynamic simulation, spring damping	7	3	3	63
Control algorithms	Timing errors, software bugs	Shuffle errors, crashes	Poor code design, insufficient testing	Software verification, redundancy	9	3	4	108
Material selection	Wear, friction issues	Premature wear, noise	Wrong plastic/metal choices	strength analysis, testing	7	2	3	42
Safety features	Guards missing or ineffective	Injury from moving parts	Design oversight	Safety interlocks, guards, standards	10	2	2	40
Human interface	Poor ergonomics, confusing controls	Operator fatigue, misuse	Poor UI layout, lack of instructions	Usability testing, labeling	7	3	3	63
Tolerances	Parts do not fit properly	Jamming, poor performance	Machining or assembly errors	Tolerance analysis, quality checks	8	4	4	128
Performance prediction	Not meeting shuffle speed or accuracy	Customer dissatisfaction, returns	Overestimated capability	Prototyping, benchmarking	8	3	3	72

Appendix N: Gantt Chart:

GANTT CHART

[illegible]

Addendum

Global Issues—Japan

If our project were to be marketed and produced in Japan, the ordering of our customer requirements by importance would see a significant change. Currently, our project focuses on card safety being first, then reliability and durability, then adjustability, speed, and cost. In Japan, one very important aspect of designing products that will be popular is reliability. Products in Japan are often designed to last a lifetime, built with high precision engineering with no faults. We currently have reliability and durability at a 9 in terms of importance, which would definitely need to be a 10 in Japan. Another aspect Japanese people value a lot is aesthetics which also includes miniaturization. Japanese consumer products always prioritize clean aesthetics and ergonomic design. In our house of quality we listed appealing design and compactness both with an importance of 6/10. These would most likely be at least 8/10 for our product to gain any traction in the Japanese market. One requirement that would decrease in importance on the Japanese market is the adjustability. In Japan, there exist gadgets for everything imaginable. Many tech products on the market exist to do one very specific thing only. For example, handheld fans which can be bought in any tech store, pocket sized bands that you can attach around your neck to keep yourself cool, or even a pen-sized breath odor checker. Our card shuffler in Japan would not require the additional functionality of being able to accommodate cards of many various dimensions.

Our card shuffler project would likely be extremely popular in Japan. This is because trading card games are extremely popular in Japan, who have created 2 of the 3 biggest trading card games of all time (Pokemon and Yugioh). There would be a much larger demand for the product in Japan compared to the US, although the room for potential growth could be a lot smaller. Although Japan is known for its love of TCGs, gambling is illegal, so the card shuffler would be unable to break into the casino market like it could in the US. There could also be some potential resistance against the product as Japan embraces tradition. Many players may enjoy the hand shuffling experience and would be unwilling to switch to an automated system. There would end up being a 50/50 split on the proportion of the target audience who would prefer the shuffler vs without.

Although the mechanical components of the product are unlikely to see any changes in price, electrical components in Japan are known for being very cheap. This would drive down our costs by a small fraction, but it would still fail to address our biggest problem in the costs which stem from the Mechanical Components.

Our product would meet the safety regulations in Japan, as they are similar to the US, but more work would be needed to ensure we met the environmental regulations. Japan is an incredibly environmentally friendly country and is very strict on packaging waste and recyclability. We would need to put additional consideration into the packaging of the product compared to simply selling it in the US

